

Legal and Regulatory Tech in Practice

Lessons learned from **TH20ECD**



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Chief Technology Officer

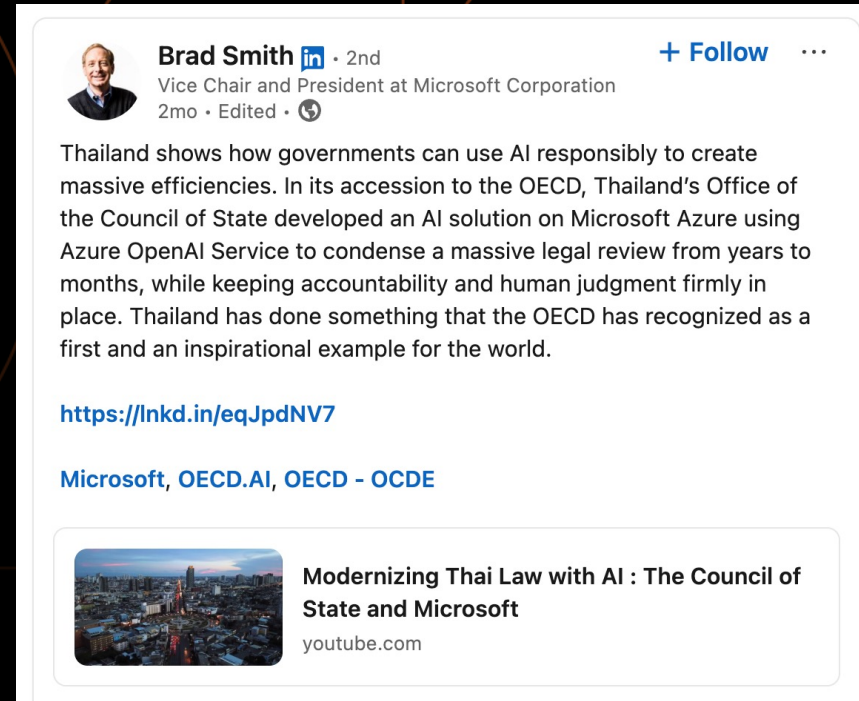


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Who we are?

Microsoft AI Innovation Partner of the year 2024

- **Thailand's Leader in Data/AI Solutions:** With unmatched expertise in Microsoft Azure, we were **the only local company** selected to showcase **three** projects at **Microsoft AI Day**, hosted by Satya Nadella in Bangkok, underscoring our leadership and innovation in the field.
- **Proven Success:** We have a strong track record of successful Data/AI implementations across **various industries**, including manufacturing, retail, compliance, legal, telecommunications, energy, banking, and cybersecurity, demonstrating our capability to deliver impactful and reliable solutions.
- **Innovative Thought Leader:** We pioneer cutting-edge technologies like self-service analytics, RAG, Graph, Knowledge Graph, GraphRAG, and Agentic Workflow. Our proven track record of introducing **valuable innovations showcases** our ability to consistently lead with impactful solutions.
- **Strong and Dedicated Team:** Our team brings a **wide range of experience**, from academic backgrounds to consulting experience.



Brad Smith  • 2nd
Vice Chair and President at Microsoft Corporation
2mo • Edited • 

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Thailand shows how governments can use AI responsibly to create massive efficiencies. In its accession to the OECD, Thailand's Office of the Council of State developed an AI solution on Microsoft Azure using Azure OpenAI Service to condense a massive legal review from years to months, while keeping accountability and human judgment firmly in place. Thailand has done something that the OECD has recognized as a first and an inspirational example for the world.

<https://lnkd.in/eJpdNV7>

Microsoft, OECD.AI, OECD - OCDE

 **Modernizing Thai Law with AI : The Council of State and Microsoft**
youtube.com



Thailand to OECD

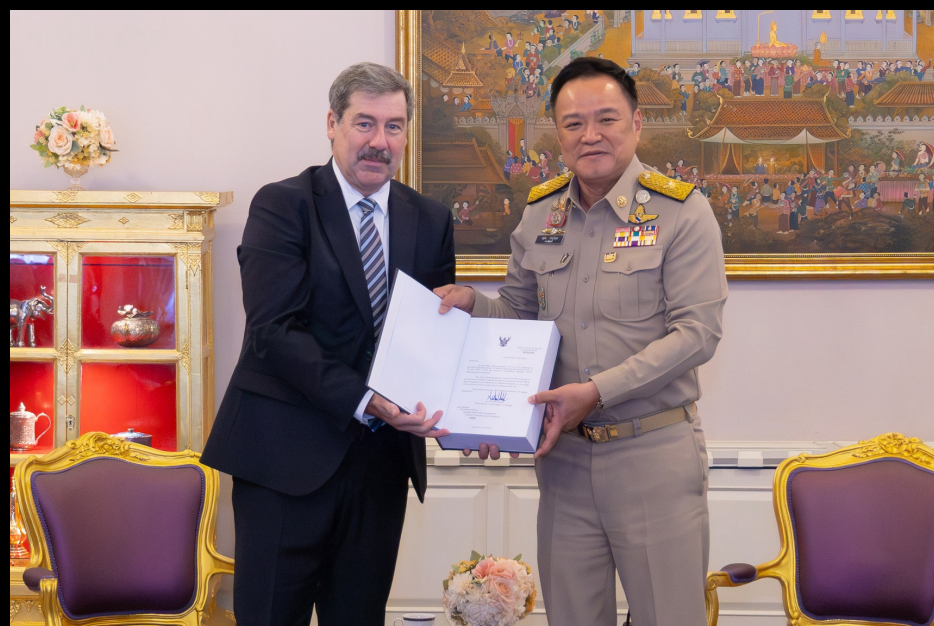
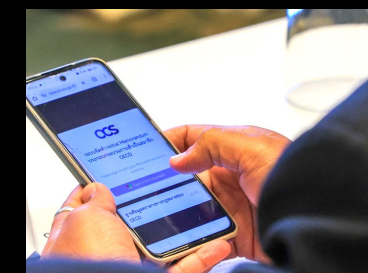
<https://inter.nesdc.go.th/th2oecd/>

Legal Intelligence Accelerates Thailand's OECD Accession Journey

Thailand's accession to the OECD is a nationally significant step toward global economic integration. Yet the process is highly complex, requiring coordination across **312 government agencies**, review of **273 legal instruments**, and processing of **70,000+ legal documents**—a task that could take decades using traditional methods.

To overcome this, we partnered with the **Office of the Council of State (OCS)** to deploy a **Generative AI-powered Legal Intelligence System**, leveraging **GraphRAG** and **Agentic AI** concepts. The system analyzes **over 1 million legal provisions**, automates compliance mapping, and provides real-time tracking, gap analysis, and reporting with exceptional speed and precision.

This modernized approach is accelerating Thailand's OECD accession, minimizing manual effort, and supporting the country's alignment with international governance standards.



TH2OECD: Accelerate Thailand's accession to OECD

Thailand's path to OECD accession is challenged by a fundamental language divide: OECD instruments are in English while Thailand's vast legal corpus exists in Thai. A single OECD standard can connect to numerous Thai laws, making simple translation inadequate and requiring broad, multi-document analysis to ensure accurate alignment.

Reform is further complicated by Thailand's layered legal structure, where primary legislation, subordinate regulations, ministerial notifications, and agency guidelines are deeply interconnected. Changing one provision often triggers cascading adjustments across many others. These linguistic, structural, and relational complexities make traditional legal review slow and highly resource-intensive—demanding a more modern, scalable approach.

70,000+

Thai Legal
Documents
Processed

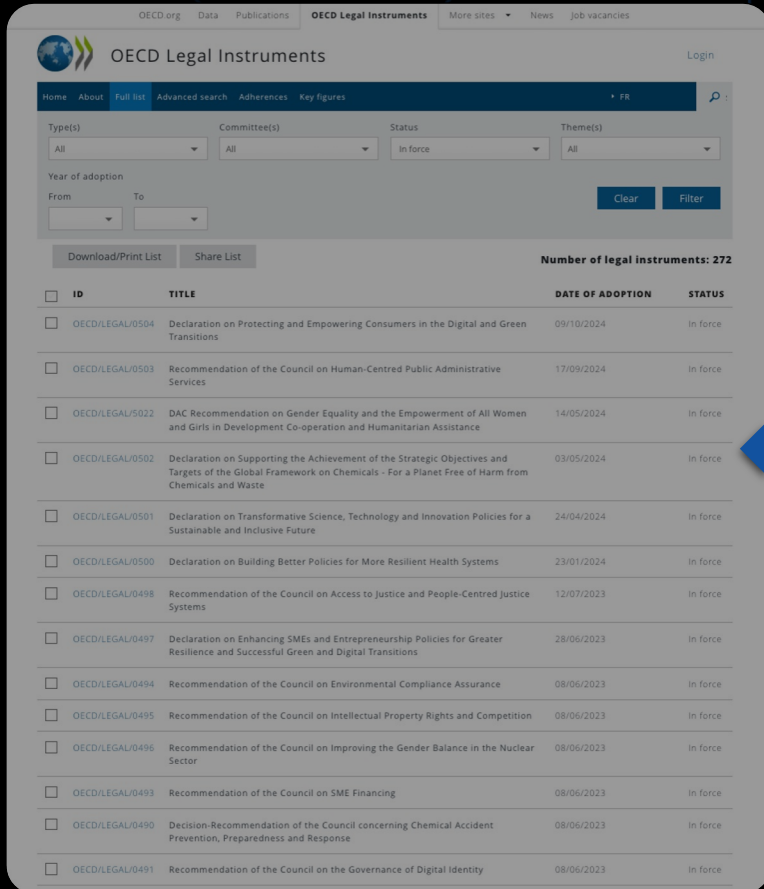
3,000,000+

Analyzed Sections
across 312
government agencies

11x Faster

Accelerate 7-10 years
of manual work into
just 11 months

How do we process this complexity in time?



ID	TITLE	DATE OF ADOPTION	STATUS
OECD/LEGAL/0504	Declaration on Protecting and Empowering Consumers in the Digital and Green Transitions	09/10/2024	In force
OECD/LEGAL/0503	Recommendation of the Council on Human-Centred Public Administrative Services	17/09/2024	In force
OECD/LEGAL/0502	DAC Recommendation on Gender Equality and the Empowerment of All Women and Girls in Development Co-operation and Humanitarian Assistance	14/05/2024	In force
OECD/LEGAL/0502	Declaration on Supporting the Achievement of the Strategic Objectives and Targets of the Global Framework on Chemicals - For a Planet Free of Harm from Chemicals and Waste	03/05/2024	In force
OECD/LEGAL/0501	Declaration on Transformative Science, Technology and Innovation Policies for a Sustainable and Inclusive Future	24/04/2024	In force
OECD/LEGAL/0500	Declaration on Building Better Policies for More Resilient Health Systems	23/01/2024	In force
OECD/LEGAL/0498	Recommendation of the Council on Access to Justice and People-Centred Justice Systems	12/07/2023	In force
OECD/LEGAL/0497	Declaration on Enhancing SMEs and Entrepreneurship Policies for Greater Resilience and Successful Green and Digital Transitions	28/06/2023	In force
OECD/LEGAL/0494	Recommendation of the Council on Environmental Compliance Assurance	08/06/2023	In force
OECD/LEGAL/0495	Recommendation of the Council on Intellectual Property Rights and Competition	08/06/2023	In force
OECD/LEGAL/0496	Recommendation of the Council on Improving the Gender Balance in the Nuclear Sector	08/06/2023	In force
OECD/LEGAL/0493	Recommendation of the Council on SME Financing	08/06/2023	In force
OECD/LEGAL/0490	Decision-Recommendation of the Council concerning Chemical Accident Prevention, Preparedness and Response	08/06/2023	In force
OECD/LEGAL/0491	Recommendation of the Council on the Governance of Digital Identity	08/06/2023	In force

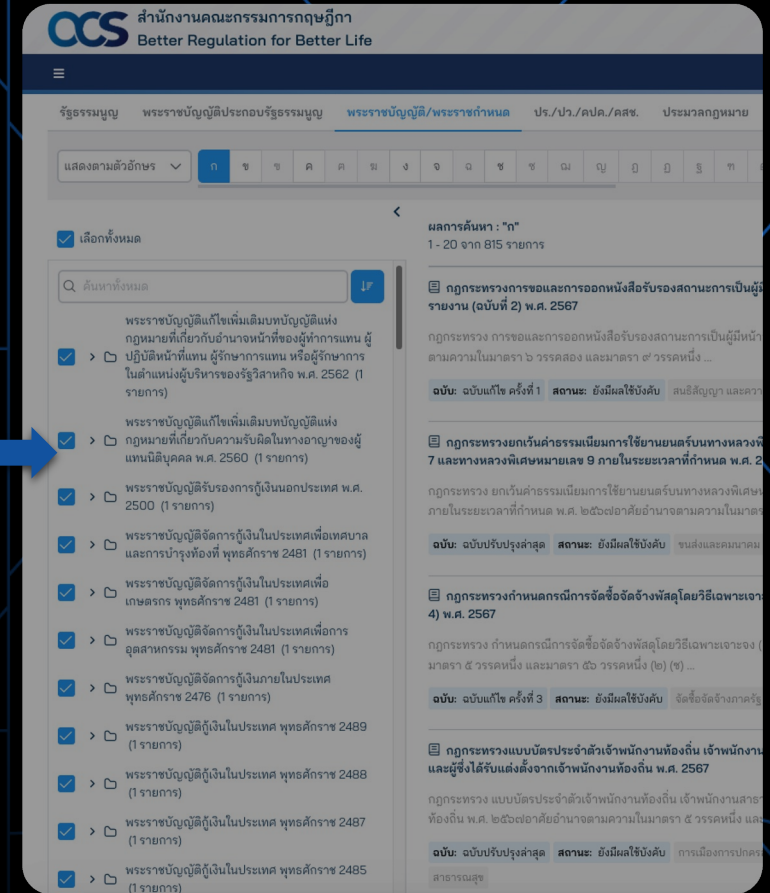


Language Gap at Scale: OECD standards are in English, while over 70,000 Thai legal documents exist only in Thai—making simple translation insufficient.

Law Identification Complexity: Before analysis can begin, relevant Thai laws, regulations, and guidelines must be accurately identified for each OECD instrument

Many-to-Many Legal Mapping: A single OECD instrument often maps to multiple Thai legal texts, across different legal tiers, requiring structured cross-document comparison at clause level.

Human-in-the-Loop at Scale: Legal alignment requires continuous expert review, validation, and accountability across shared evidence—demanding a platform built for HITL at scale, not a chatbot-centric workflow.



รัฐธรรมนูญ	พระราชบัญญัติประกอบรัฐธรรมนูญ	พระราชบัญญัติ/พระราชกำหนด	ป.ร./ป.ร./ค.บ./ค.ส.	ประมวลกฎหมาย
เลือกทั้งหมด				
ค้นหาทั้งหมด				
พระราชบัญญัติแก้ไขเพิ่มเติมบทบัญญัติแห่งกฎหมายที่เกี่ยวข้องกับอำนาจหน้าที่ของผู้บริหารการปฏิบัติหน้าที่แทน ผู้รักษาการแทน หรือผู้รักษาการในตำแหน่งผู้บริหารของรัฐบาล พ.ศ. 2562 (1 รายการ)				
พระราชบัญญัติแก้ไขเพิ่มเติมบทบัญญัติแห่งกฎหมายที่เกี่ยวข้องกับความรับผิดชอบในทางอาญาของผู้แทนนิติบุคคล พ.ศ. 2560 (1 รายการ)				
พระราชบัญญัติรับรองการกู้เงินนอกประเทศ พ.ศ. 2500 (1 รายการ)				
พระราชบัญญัติจัดการกู้เงินในประเทศเพื่อเทศบาล และการบำรุงท้องที่ พุทธศักราช 2481 (1 รายการ)				
พระราชบัญญัติจัดการกู้เงินในประเทศเพื่อเกษตรกรรม พุทธศักราช 2481 (1 รายการ)				
พระราชบัญญัติจัดการกู้เงินในประเทศเพื่อการอุตสาหกรรม พุทธศักราช 2481 (1 รายการ)				
พระราชบัญญัติจัดการกู้เงินภายในประเทศ พุทธศักราช 2476 (1 รายการ)				
พระราชบัญญัติกู้เงินในประเทศ พุทธศักราช 2489 (1 รายการ)				
พระราชบัญญัติกู้เงินในประเทศ พุทธศักราช 2488 (1 รายการ)				
พระราชบัญญัติกู้เงินในประเทศ พุทธศักราช 2487 (1 รายการ)				
พระราชบัญญัติกู้เงินในประเทศ พุทธศักราช 2485 (1 รายการ)				

The Design

OECD AI Principles

Values-based principles



Inclusive growth, sustainable development and well-being



Human right and democratic values, including fairness and privacy



Transparency and explainability



Robustness, security and safety



Accountability



Stakeholders should proactively engage in responsible stewardship of trustworthy AI in pursuit of beneficial outcomes for people and the planet, such as augmenting human capabilities and enhancing creativity, advancing inclusion of underrepresented populations, reducing economic, social, gender and other inequalities, and protecting natural environments, thus invigorating inclusive growth, well-being, sustainable development and environmental sustainability.

AI actors should respect the rule of law, human rights, democratic and human-centred values throughout the AI system lifecycle. These include non-discrimination and equality, freedom, dignity, autonomy of individuals, privacy and data protection, diversity, fairness, social justice, and internationally recognised labour rights. This also includes addressing misinformation and disinformation amplified by AI, while respecting freedom of expression and other rights and freedoms protected by applicable international law.

To this end, AI actors should implement mechanisms and safeguards, such as capacity for human agency and oversight, including to address risks arising from uses outside of intended purpose, intentional misuse, or unintentional misuse in a manner appropriate to the context and consistent with the state of the art.

AI Actors should commit to transparency and responsible disclosure regarding AI systems. To this end, they should provide meaningful information, appropriate to the context, and consistent with the state of art:

- to foster a general understanding of AI systems, including their capabilities and limitations,
- to make stakeholders aware of their interactions with AI systems, including in the workplace,
- where feasible and useful, to provide plain and easy-to-understand information on the sources of data/input, factors, processes and/or logic that led to the prediction, content, recommendation or decision, to enable those affected by an AI system to understand the output, and,
- to provide information that enable those adversely affected by an AI system to challenge its output.

AI systems should be robust, secure and safe throughout their entire lifecycle so that, in conditions of normal use, foreseeable use or misuse, or other adverse conditions, they function appropriately and do not pose unreasonable safety and/or security risks.

Mechanisms should be in place, as appropriate, to ensure that if AI systems risk causing undue harm or exhibit undesired behaviour, they can be overridden, repaired, and/or decommissioned safely as needed.

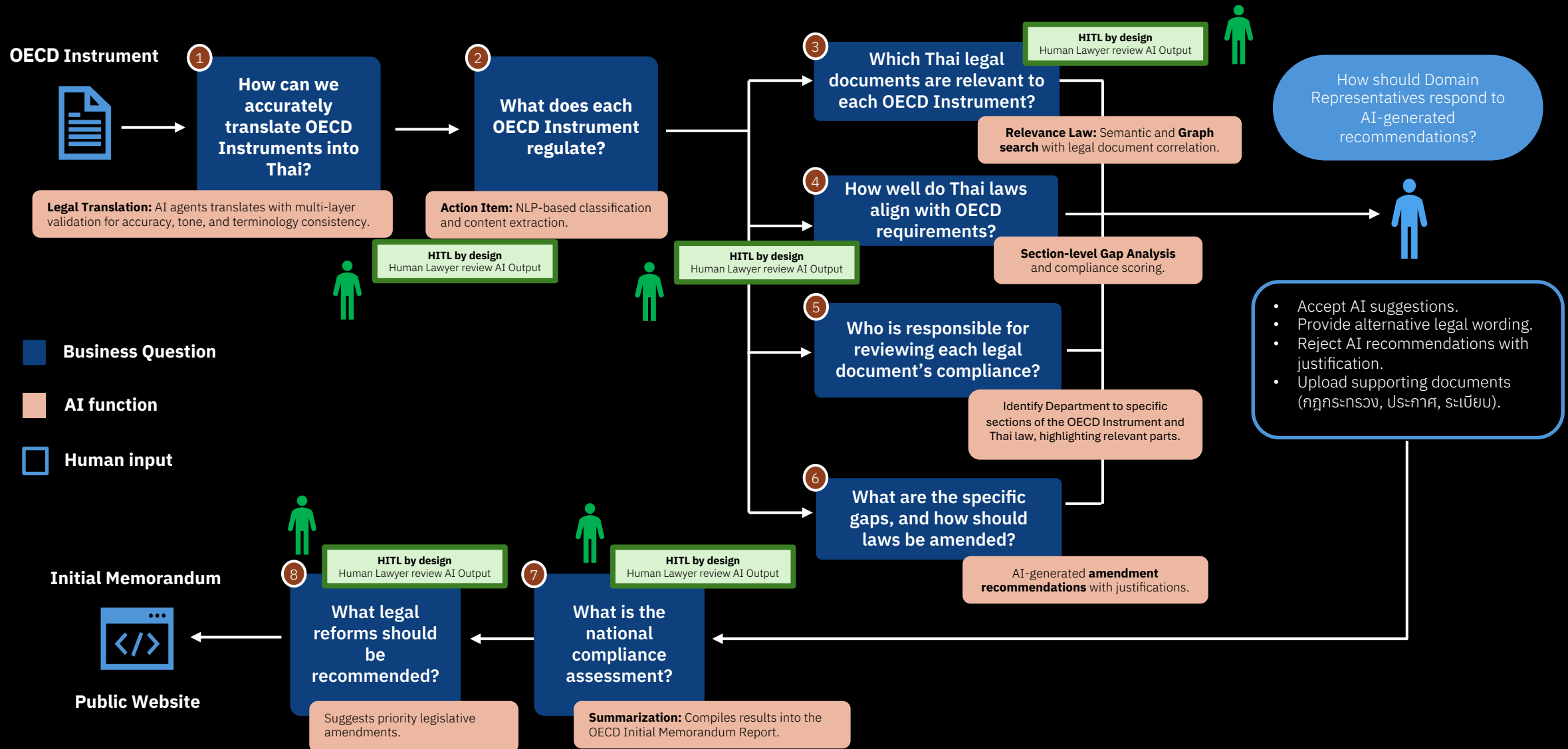
Mechanisms should also, where technically feasible, be in place to bolster information integrity while ensuring respect for freedom of expression.

AI actors should be accountable for the proper functioning of AI systems and for the respect of the above principles, based on their roles, the context, and consistent with the state of the art.

To this end, AI actors should ensure traceability, including in relation to datasets, processes and decisions made during the AI system lifecycle, to enable analysis of the AI system's outputs and responses to inquiry, appropriate to the context and consistent with the state of the art.

AI actors, should, based on their roles, the context, and their ability to act, apply a systematic risk management approach to each phase of the AI system lifecycle on an ongoing basis and adopt responsible business conduct to address risks related to AI systems, including, as appropriate, via co-operation between different AI actors, suppliers of AI knowledge and AI resources, AI system users, and other stakeholders. Risks include those related to harmful bias, human rights including safety, security, and privacy, as well as labour and intellectual property rights.

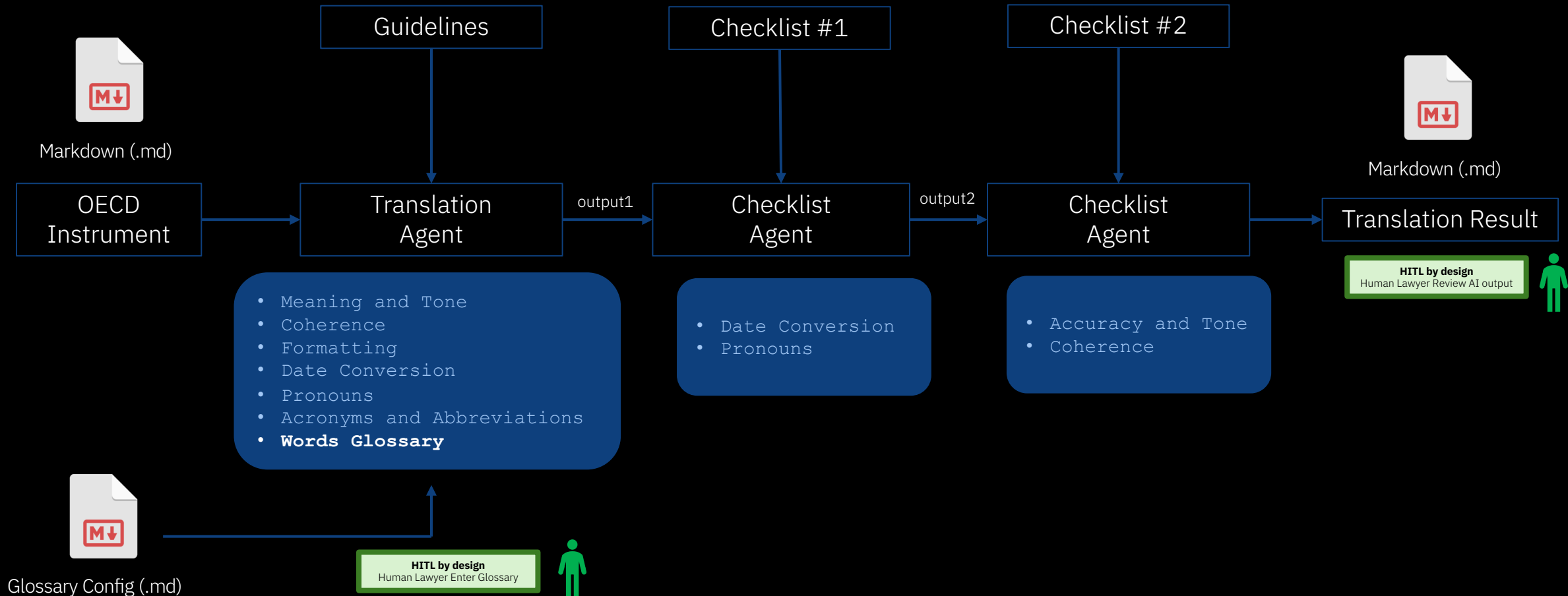
Thailand20CED: Key Questions & AI Contributions



The Translation

Agentic AI Engine : Translation

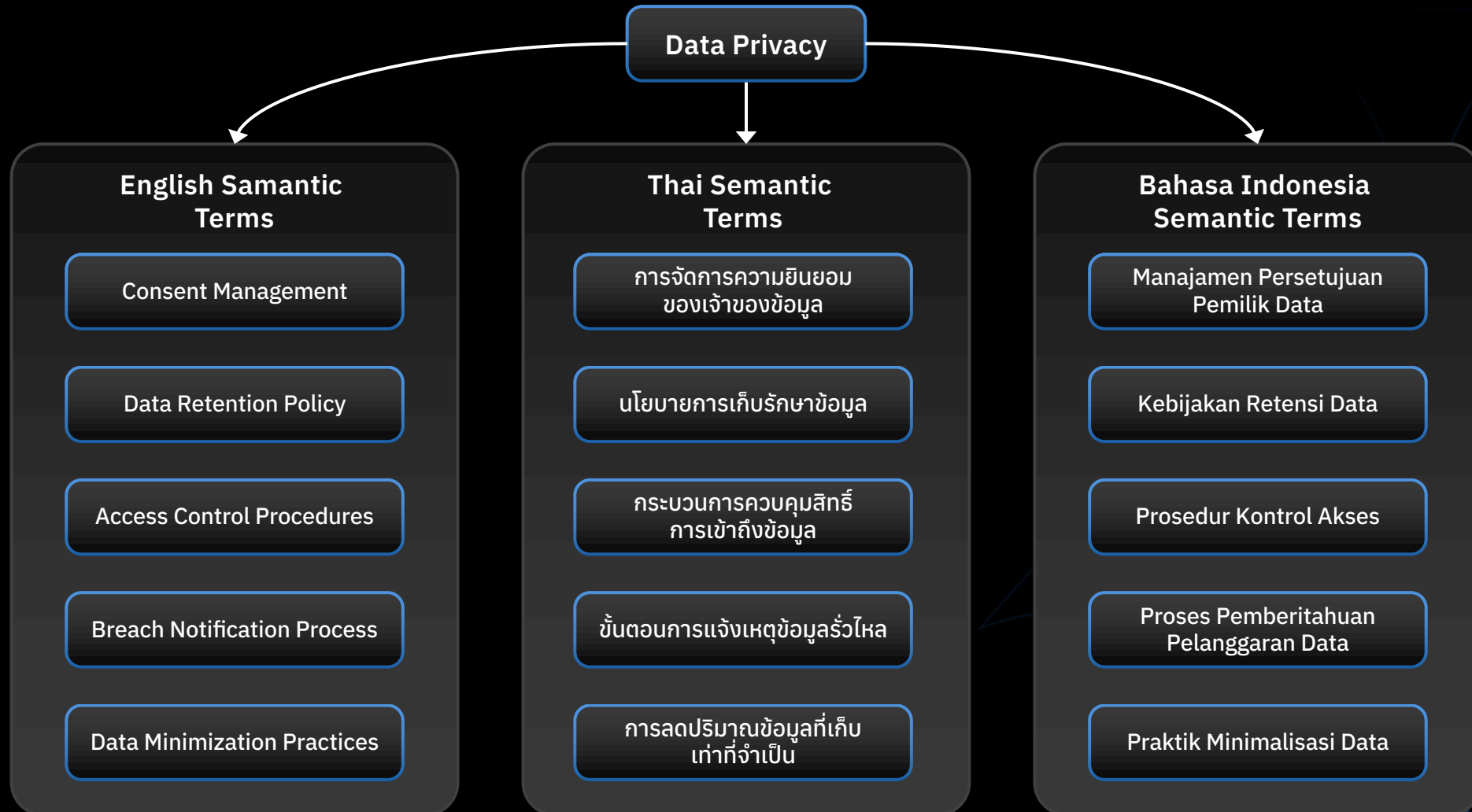
AI workflow: 3-layer sequential chain



The Selection and Mapping

Semantic matching

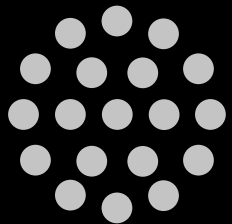
Rule-based systems match words; semantic matching connects meaning.



Getting the data right for legal document

We create relational data for legal documents (GraphRAG) to enable AI to retrieve information more accurately and comprehensively than traditional retrieval methods (RAG).

Section

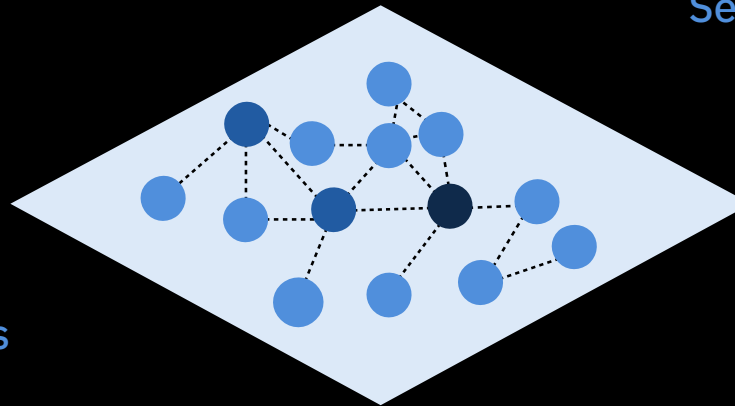


Chunk



Relationships

Legislation

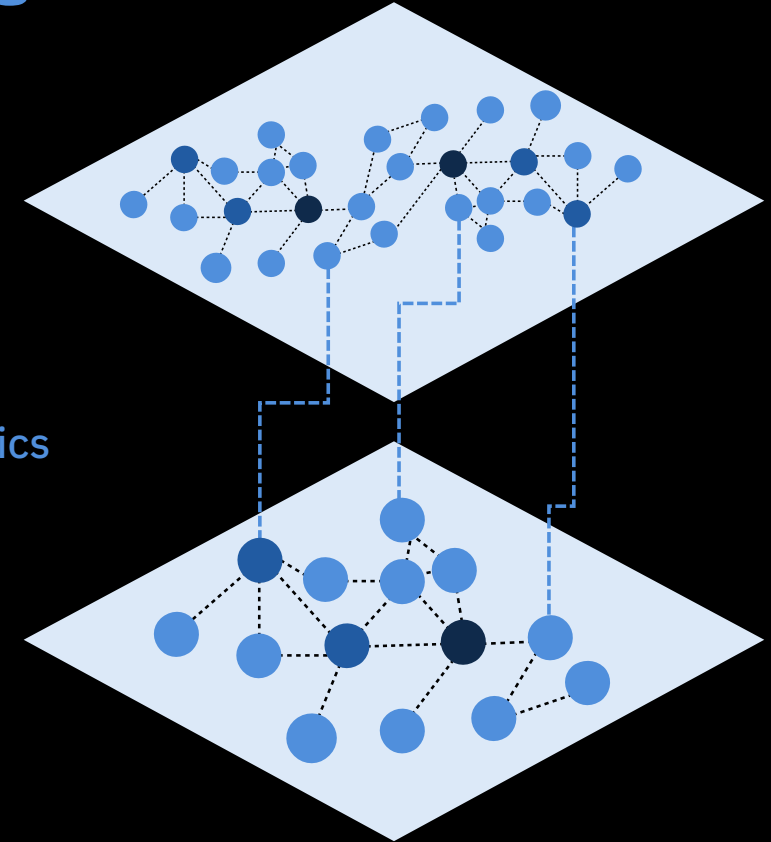


Document Relationship



Semantics

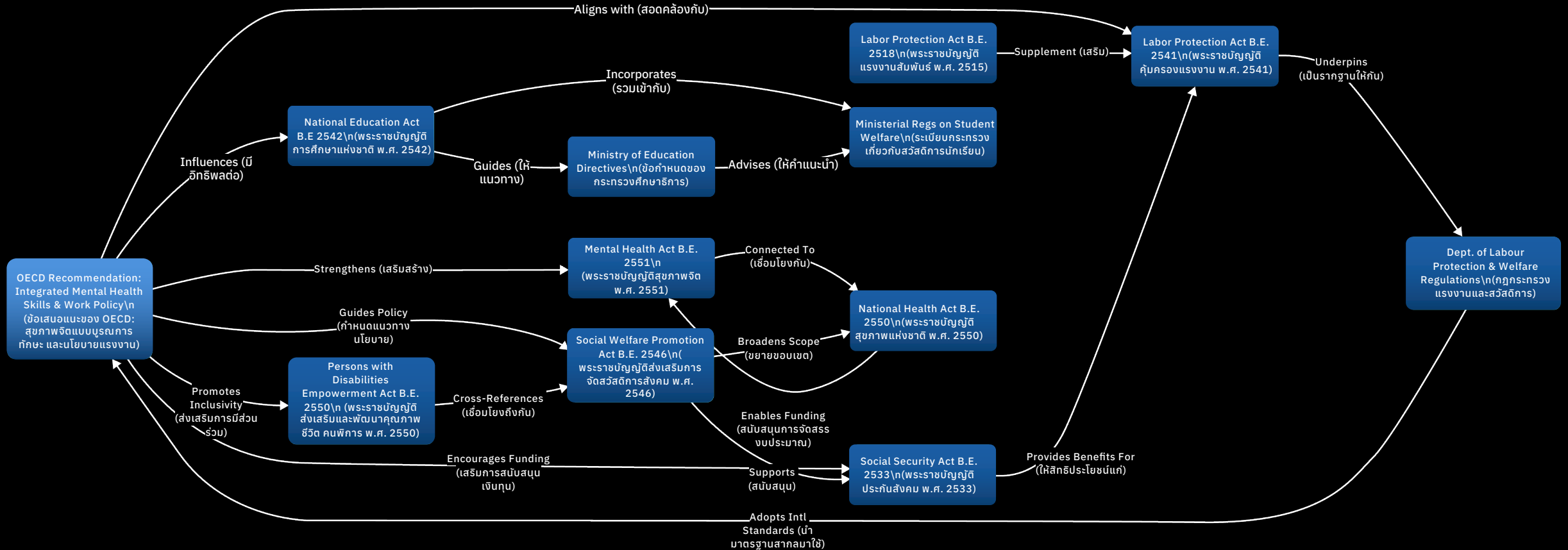
Legislation + Notification + Rules



Semantic Relationship

GraphRAG Query

Topic-Based Legal Search Across All Levels of Thai Regulatory Texts



Engineering Lesson learned

Impact = We need to think about these



Workflow and
Automation



Governance
and the Agents



Data and Trust

The “Agent”

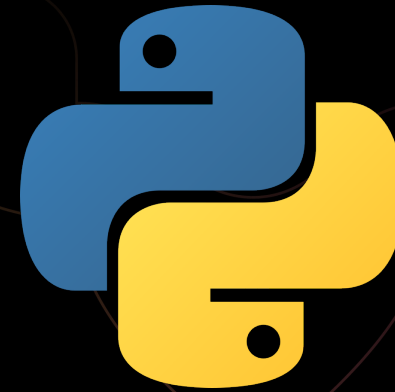


Heuristic Systems

A **heuristic system** uses rules-of-thumb, approximations, or educated guesses to produce solutions that are *good enough*, but not guaranteed optimal.

Core Characteristics

- May involve randomness
- Produces approximate solutions
- Often faster than exact methods
- Not guaranteed to produce the same result every time



Deterministic Systems

A deterministic system produces the same output for a given input every time, with no randomness or approximation.

Core Characteristics

- Fully predictable behavior
- No randomness
- Rule-based or mathematically defined
- Repeatable and reproducible results

The Governance Design Philosophy: Balance

No oversight → no friction but errors



Too much oversight → friction, no automation gain.



Principles for Building Reliable and Scalable Systems

1

Iteration as the Core Engine

Build through repeated cycles of testing, expert feedback, failure analysis, and refinement. The first version is rarely strong enough—quality improves when every expert correction becomes repeatable system behavior.

2

Observability for Faster Improvement

Track every agent output, retrieved source, intermediate result, rubric, score, and version change—so engineers can pinpoint whether failures come from retrieval, classification, reasoning, formatting, or scoring.

3

Domain-Grounded Reasoning

Design around domain rules, not keywords or text similarity. The agent must understand source hierarchy, legal authority, policy tests, and concept boundaries to distinguish terms that look alike but differ in meaning.

4

Right Boundary Between AI and Code

Use the LLM for ambiguity, reading, comparison, reasoning, and drafting. Use deterministic code for validation, required fields, scoring, version control, and output packaging—knowing when not to use the LLM.

5

Balance Generic and Specific Design

Build reusable components, but allow targeted customization where the domain demands it. Too specific won't scale; too generic misses edge cases like local document structures, repositories, date formats, and source conventions.

THANK YOU